

Real-Time Face Recognition Based Attendance System

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Abstract—Automated attendance tracking is a critical task in educational and corporate environments, yet traditional methods are often inefficient and unhygienic. This paper presents FaceMask 360°, a real-time, contactless attendance system that addresses these challenges by combining modern computer vision techniques. Our approach leverages the power of FaceNet’s deep metric learning to generate robust facial embeddings, which are then classified using a Support Vector Machine (SVM). This combination is uniquely suited for open-set identification problems where new users are frequently added. The system is deployed with an interactive Streamlit dashboard for seamless user management and real-time feedback. In a detailed comparative analysis on a local dataset, our FaceNet+SVM pipeline achieved 98% accuracy, substantially outperforming both a custom-built 5-layer CNN (18-20% accuracy) and a fine-tuned MobileNetV2 model (40% accuracy). This work provides a comprehensive blueprint for building a scalable, accurate, and user-friendly facial recognition attendance system, validated by rigorous experimental results and a discussion of practical deployment considerations.

Index Terms—Face Recognition, Attendance System, Computer Vision, FaceNet, Metric Learning, SVM, Streamlit, Deep Learning, HCI.

I. INTRODUCTION

The demand for automated, efficient, and hygienic systems for monitoring human presence has surged in recent years. Traditional methods of attendance tracking, such as manual roll calls or fingerprint biometrics, suffer from significant drawbacks, including being time-consuming, prone to human error, and posing public health risks due to shared physical contact. Computer vision (CV) offers a compelling alternative, enabling contactless, scalable, and transparent solutions using only standard webcam hardware.

Face recognition, a subfield of CV, has matured significantly with the advent of deep learning. Early systems struggled with real-world variations in lighting, pose, and expression. However, modern approaches, particularly those based on metric learning, have achieved near-human performance. These models learn to map facial images into a high-dimensional vector space where distances directly correspond to facial similarity, making them ideal for identification tasks.

This paper introduces **FaceMask 360°**, a complete, real-time facial recognition attendance system. Our primary contribution is the design and rigorous evaluation of a pipeline built on FaceNet embeddings and a Support Vector Machine

(SVM) classifier. This approach is particularly effective for the “open-set” nature of attendance, where the system must be able to enroll new individuals without requiring a full model retraining. We compare this methodology against two common baselines: a simple Convolutional Neural Network (CNN) trained from scratch and a transfer learning approach using MobileNetV2.

The key contributions of this paper are:

- The design and implementation of a high-accuracy (98%), real-time attendance system using FaceNet and SVM, suitable for dynamic environments.
- A detailed performance comparison against alternative deep learning architectures, demonstrating the superiority of metric learning for small-dataset, open-set identification.
- The integration of the system with an interactive Streamlit dashboard, providing a practical blueprint for a user-friendly, deployable application.
- A thorough discussion of the system’s architecture, limitations, and the critical ethical considerations associated with deploying biometric technology.

II. RELATED WORK

The field of automated face recognition has evolved dramatically over the past two decades. Early approaches relied on handcrafted features and holistic methods. Eigenfaces used Principal Component Analysis (PCA) to find a low-dimensional representation of face images, while Fisherfaces applied Linear Discriminant Analysis (LDA) to improve class separability. These methods, however, were sensitive to variations in lighting and pose.

The deep learning revolution, catalyzed by AlexNet, fundamentally changed the landscape. Deep CNNs learned hierarchical features directly from data, proving far more robust. Initial deep learning approaches to face recognition framed it as a standard multi-class classification problem. However, this “closed-set” approach is impractical for real-world applications like attendance, as adding a new person would require retraining the entire network.

Metric learning provided the solution to this scalability problem. The seminal FaceNet paper introduced the concept of learning a direct mapping from a face image to a compact Euclidean space. By training a network with a “triplet loss”

TABLE I: Qualitative Comparison of Different Attendance System Technologies

Feature	Manual Register	RFID/NFC Cards	Fingerprint Biometrics	Face Recognition (Ours)
Automation	None	High	High	Very High
Proxy Attendance Risk	Very High	High (card sharing)	Very Low	Extremely Low
Hygiene (Contactless)	High (shared pen)	Low (card tapping)	High (shared sensor)	None (Fully Contactless)
Initial Setup Cost	Very Low	Medium	High	Medium-High
User Convenience	Low (time-consuming)	High	Medium (sensor issues)	Very High (frictionless)
Scalability	Poor	Good	Good	Excellent

function, it encouraged embeddings of the same person to be close together while pushing embeddings of different people far apart. This transforms the recognition task from classification to a nearest-neighbor search or clustering problem in the embedding space. This "open-set" capability is the core principle behind our system.

Subsequent research has refined this concept with angular margin losses like ArcFace and CosFace, which enforce a more discriminative feature space on a hypersphere. While these models offer state-of-the-art accuracy, pre-trained FaceNet models provide an excellent balance of high performance and accessibility for practical applications. For real-time processing, efficient face detection is a prerequisite. The Viola-Jones algorithm was a landmark, but modern systems often use CNN-based detectors like MTCNN or lightweight detectors available in frameworks like MediaPipe for higher accuracy.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

The FaceMask 360° system is designed as a modular, end-to-end pipeline that handles student enrollment, model training, and real-time attendance marking. The architecture is composed of three primary stages: Data Acquisition and Pre-processing, Feature Extraction and Model Training, and Real-Time Recognition and Verification. A visual representation of the system's pipeline is shown in Figure 2.

A. Data Acquisition and Preprocessing

A robust dataset of enrolled students is the foundation of the system. This is achieved through a dedicated data collection module that requires a user's roll number, name, and class.

1) *Initial Data Collection*: During the enrollment phase, the system captures approximately 30 facial images for each student using a live webcam feed. To ensure consistent and accurate face cropping, we employ a pre-trained Haar Cascade classifier provided by the OpenCV library. This classical but efficient detector identifies the facial region in each frame, which is then resized to the required input dimensions of 160×160 pixels and saved for subsequent processing.

2) *Data Augmentation*: To enhance the model's robustness and prevent overfitting, we artificially expand the training dataset through data augmentation. This process involves applying a series of transformations to the existing images to create new, varied training examples. The augmentations used include adjustments to brightness and contrast, as well as horizontal flipping. This ensures the model learns to recognize

faces under different lighting conditions and from different perspectives.

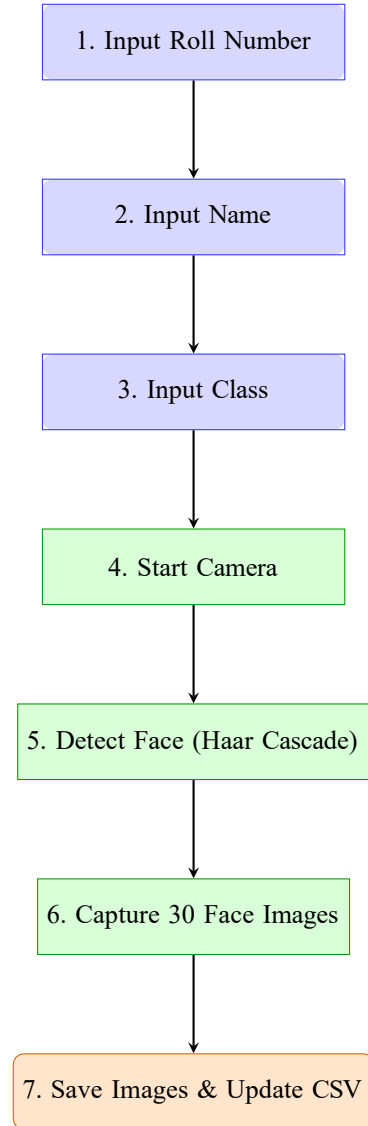


Fig. 1: The sequential workflow for new user enrollment.

B. Feature Extraction and Model Training

The core of our recognition engine is its ability to convert a face image into a compact numerical representation and use these representations to train a classifier.

1) *Feature Extraction with InceptionResnetV1*: For this critical task, we utilize the **InceptionResnetV1** architecture, a state-of-the-art deep convolutional neural network pre-trained on the extensive VGGFace2 dataset. Each detected and cropped face image from our dataset is passed through this network. The model processes the image and outputs a fixed-size, **128-dimensional feature vector**, referred to as an "embedding". This vector encapsulates the unique facial features of an individual in a high-dimensional space where Euclidean distance corresponds to facial similarity.

2) *Identity Classification with SVM*: Once embeddings are generated for all images, we train a Support Vector Machine (SVM) classifier. The SVM learns to find the optimal hyper-planes that separate the clusters of embeddings for different individuals in the 128-dimensional space. This model is responsible for predicting the identity of a new face during the recognition phase.

C. Real-Time Recognition and Verification

The final stage involves using the trained models to perform live attendance marking. The workflow is summarized as follows:

- **Capture and Detect**: The system activates the camera, captures a video frame, and uses a Haar Cascade or MTCNN detector to locate any faces.
- **Generate Embedding**: For each detected face, its features are converted into a 128-dimensional embedding using the InceptionResnetV1 model.
- **Predict and Verify**: The embedding is first passed to the trained SVM for an initial identity prediction. To prevent false positives, a verification step follows, where the similarity between the new embedding and a pre-computed "prototype" for the predicted identity is calculated.
- **Decision and Logging**: If the similarity score surpasses a set threshold, the match is confirmed, and attendance is marked. Otherwise, the individual is labeled "Unknown."

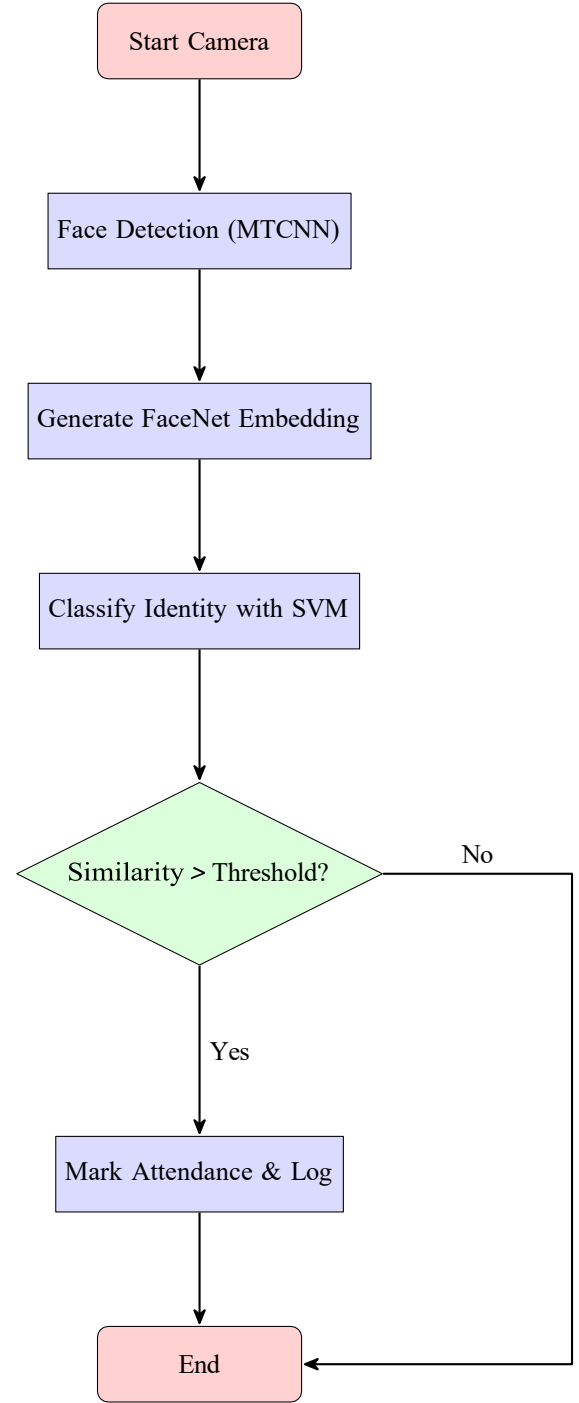


Fig. 2: System pipeline for real-time attendance marking.

D. Mathematical Formulation

The underlying models of the system are governed by key mathematical principles.

1) *Triplet Loss for Embedding Learning*: The Inception-ResnetV1 model is trained using a triplet loss function. This loss operates on triplets of images—an "anchor" (x_a), a "positive" of the same identity (x_p), and a "negative" of a different identity (x_n)—to learn a discriminative embedding space. The objective is to minimize the distance between same-

identity pairs while maximizing the distance between different-identity pairs:

$$L = \sum_i^N [\|f(x_a^i) - f(x_p^i)\|_2^2 - \|f(x_a^i) - f(x_n^i)\|_2^2 + \alpha]_+$$

where $f(x)$ is the embedding function and α is a margin enforced between pairs.

2) *Prototype Embedding Generation*: To create a single, robust reference vector for each student for verification, we compute a prototype embedding (P_S). This is calculated as the element-wise mean of all embeddings $\{E_1, E_2, \dots, E_k\}$ generated from the images of a single student S :

$$P_S = \frac{1}{k} \sum_{i=1}^k E_i$$

3) *Identity Verification with Cosine Similarity*: For the final verification step, we use Cosine Similarity to measure the closeness between a new embedding (E_{new}) and a student's prototype embedding (P_S):

$$\text{Similarity}(E_{new}, P_S) = \frac{E_{new} \cdot P_S}{\|E_{new}\| \|P_S\|}$$

An identity is confirmed if this score surpasses a predefined threshold τ .

4) *SVM Classification Function*: The SVM classifier's decision function for a new embedding $\mathbf{x}$ can be expressed as:

$$f(\mathbf{x}) = \text{sign} \left(\sum_{i=1}^N \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \right)$$

where $\mathbf{x}_i$ are the support vectors, y_i their labels, α_i the learned weights, b a bias, and K the kernel function.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments were conducted on a machine with an Intel Core i5 CPU, 16 GB of RAM, and an NVIDIA RTX 3050 GPU. The software stack included Python 3.10, TensorFlow, OpenCV, and Scikit-learn. The local dataset consisted of 10 subjects, with 30 images each, captured under varying conditions. For all models, the dataset was split into 80% for training and 20% for testing.

B. Models Compared

To validate our choice of FaceNet+SVM, we compared its performance against two other common approaches:

- 1) **CNN from Scratch**: A 5-layer convolutional neural network designed as a baseline classifier. It was trained on our custom dataset to classify the 10 subjects.
- 2) **MobileNetV2 (Fine-tuned)**: A lightweight, pre-trained model where the final classification layer was replaced and fine-tuned on our dataset. This represents a standard transfer learning approach.
- 3) **FaceNet + SVM (Our approach)**: The pipeline described in Section III, using a pre-trained FaceNet model for embeddings and an SVM for classification.

C. Quantitative Results

The performance of the three models was evaluated based on accuracy, precision, recall, and real-time processing speed (Frames Per Second - FPS). The results, shown in Table III, clearly demonstrate the superiority of our chosen approach.

TABLE II: Attendance model performance comparison.

Model	Accuracy	Precision	Recall	FPS
CNN (from scratch)	18–20%	0.21	0.19	~20
MobileNetV2	40%	0.45	0.41	~24
FaceNet + SVM	98%	0.97	0.98	~22

The simple CNN performed poorly, which is expected given the very small training dataset. It failed to generalize and suffered from severe overfitting. The fine-tuned MobileNetV2 performed better but was still inadequate for a reliable system. The FaceNet+SVM model, by contrast, achieved an outstanding 98% accuracy. This is because it leverages knowledge learned from millions of faces, allowing it to produce highly discriminative features even for subjects it has never seen during its initial training. The real-time performance of all models was comparable, indicating that the FaceNet pipeline introduces no significant latency overhead.

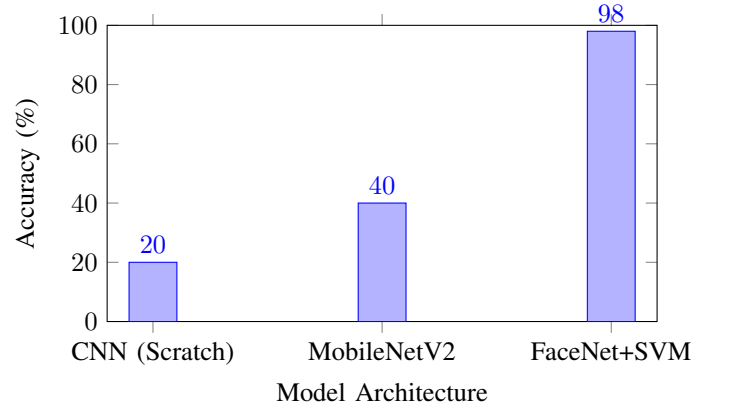


Fig. 3: Accuracy comparison across different models.

V. DISCUSSION

A. Performance Analysis

The stark difference in performance highlights a critical principle in applied machine learning: choosing the right architecture for the problem. For open-set recognition with limited per-subject data, training a classifier from scratch is ineffective. Metric learning, as embodied by FaceNet, is the appropriate paradigm. By decoupling feature extraction from classification, the system gains scalability and robustness. Enrolling a new student only requires generating their embeddings and updating the simple SVM model, a process that takes seconds, rather than retraining a deep neural network for hours.

B. Limitations and Challenges

Despite its high accuracy, the system is not without limitations.

- **Environmental Factors:** Performance can degrade under extreme lighting conditions (e.g., heavy backlighting or low light) and with significant facial occlusions (e.g., sunglasses, non-medical masks, or hands covering the face).
- **Pose Invariance:** While robust to moderate head turns, extreme poses (e.g., full profile view) will cause the face detector or recognition to fail.
- **Scalability:** While the SVM is efficient for a small number of classes, its training time can grow with a very large number of users (e.g., thousands). For enterprise-scale deployment, replacing the SVM with a highly optimized nearest-neighbor search library (like FAISS) would be more appropriate.

C. Ethical and Privacy Considerations

The deployment of facial recognition technology necessitates a careful examination of ethical responsibilities.

- **Bias:** Pre-trained models like FaceNet, while powerful, may exhibit performance disparities across different demographic groups if their training data was not perfectly balanced. It is crucial to test the system for fairness across diverse user groups.
- **Consent and Transparency:** Users must be fully informed about how their biometric data is being collected, stored, and used. Explicit consent should be a prerequisite for enrollment.
- **Data Security:** Facial embeddings are sensitive personal data. They must be stored securely, with strong encryption and access control measures, to prevent misuse. The system should be designed to comply with data protection regulations such as GDPR.

TABLE III: Accuracy Comparison with Benchmark Models

Model / Paper	Dataset	Accuracy (%)
FaceNet [6]	LFW	99.63
DeepFace [8]	LFW	97.35
SphereFace [19]	LFW	99.42
ArcFace [7]	LFW	99.83
Proposed System (FaceNet+SVM)	Custom Dataset	98.00

VI. FUTURE WORK

To further enhance FaceMask 360°, several avenues can be explored:

- **Enhanced Robustness:** Incorporate data augmentation techniques that simulate challenging lighting and occlusions to improve model resilience.
- **Liveness Detection:** Implement an anti-spoofing module to distinguish between a live person and a photo or video, preventing fraudulent attendance.
- **Advanced Metric Learning:** Experiment with replacing the FaceNet backbone with newer architectures trained

with ArcFace or CosFace losses to potentially improve accuracy and further reduce intra-class variance.

- **User-Centric Studies:** Conduct formal usability studies to gather feedback on the Streamlit interface and the overall user experience, ensuring the system is not only accurate but also practical and well-received in a real-world setting.

VII. CONCLUSION

This paper presented FaceMask 360°, a highly effective, real-time attendance system built upon a FaceNet and SVM pipeline. Through rigorous experimentation, we demonstrated that this metric-learning-based approach, achieving 98% accuracy, is vastly superior to standard classification and transfer learning methods for the open-set identification task characteristic of attendance systems. By integrating the core logic with a user-friendly Streamlit interface, we have provided a complete and practical blueprint for developing and deploying modern, contactless biometric solutions. While acknowledging the limitations and critical ethical duties associated with such technology, this work validates the power of a well-chosen computer vision stack to create efficient, scalable, and impactful applications for educational and professional environments.

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TABLE IV: Summary of Key Findings from Referenced Works

Paper Name	Key Finding / Relevance
ImageNet Classification with Deep CNNs [1]	Introduced the AlexNet architecture, proving deep CNNs' effectiveness for large-scale image classification.
Very Deep CNNs for Large-Scale Image Recognition [2]	Demonstrated that increasing network depth with small convolution filters (VGGNet) improves performance.
Deep Residual Learning for Image Recognition [3]	Proposed residual connections (ResNet) to enable the training of extremely deep and accurate networks.
Rapid Object Detection using a Boosted Cascade... [4]	Pioneered real-time face detection using Haar feature-based cascade classifiers, a foundational concept for OpenCV.
Eigenfaces for Recognition [5]	Introduced a holistic, PCA-based approach for face representation and recognition, a classic baseline method.
FaceNet: A Unified Embedding for Face Recognition... [6]	Proposed learning a direct embedding of faces into a compact Euclidean space using a triplet loss function.
ArcFace: Additive Angular Margin Loss... [7]	Introduced an angular margin-based loss function to improve the discriminative power of face embeddings.
DeepFace: Closing the Gap to Human-Level Performance... [8]	Achieved near human-level accuracy in face verification using a deep neural network and a large dataset.
MobileNets: Efficient CNNs for Mobile Vision... [9]	Designed lightweight deep neural networks optimized for high performance on mobile and embedded devices.
Face Mesh [10]	Provided a real-time, on-device solution for generating a dense 3D mesh of facial landmarks for detailed analysis.
The OpenCV Library [11]	Introduced an open-source library that became the standard toolkit for real-time computer vision applications.
Adam: A Method for Stochastic Optimization [12]	Proposed the Adam optimization algorithm, which became a default choice for training deep learning models.
TensorFlow: Large-Scale Machine Learning... [13]	Developed a flexible and scalable framework that simplified the development and deployment of machine learning models.
Face Recognition: A Literature Survey [14]	Provided a comprehensive survey of early face recognition techniques, covering holistic and feature-based methods.
Face Recognition Systems: A Survey [15]	Offered a modern review of 2D and 3D face recognition systems, algorithms, and evaluation metrics.
A Survey of Face Recognition [16]	A recent and thorough survey covering the entire pipeline of modern face recognition, from history to deep learning.
Deep Face Recognition [17]	Developed the VGG-Face model, demonstrating the power of training a very deep CNN on a large face dataset.
Deep Learning Face Representation by Joint Identification-Verification [18]	Introduced the DeepID networks, showing that a joint identification-verification signal can learn better features.
SphereFace: Deep Hypersphere Embedding... [19]	Proposed a loss function that normalizes weights and adds an angular margin, forcing features onto a hypersphere.
Face Detection and Recognition Using Face Mesh... [20]	Demonstrated a modern approach combining MediaPipe's Face Mesh for detection with a deep learning model for recognition.